

# Erdős Problem #561

EP-561

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

## ORIGINAL PROBLEM STATEMENT

Let  $\hat{R}(G)$  denote the size Ramsey number, the minimal number of edges  $m$  such that there is a graph  $H$  with  $m$  edges such that in any 2-colouring of the edges of  $H$  there is a monochromatic copy of  $G$ .

Let  $F_1$  and  $F_2$  be the union of stars. More precisely, let  $F_1 = \cup_{i \leq s} K_{1, n_i}$  and  $F_2 = \cup_{j \leq t} K_{1, m_j}$ . Prove that  $\hat{R}(F_1, F_2) = \sum_{2 \leq k \leq s+t} \max\{n_i + m_j - 1 : i + j = k\}$ .

**Source and attribution.** Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>